MCP

(Model Context Protocol)

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📚 What is the Model Context Protocol (MCP)?

> The **Model Context Protocol (MCP)** is a standard that defines how **applications** (clients) and **servers** provide **contextual data, tools, and workflows** to **LLMs (Large Language Models)**.

Think of MCP like the **"USB for LLMs":** Just like USB standardizes how devices connect, MCP standardizes how **tools**, **resources**, and **prompts** connect to LLMs.

WHY DOES MCP MATTER?

`Without MCP`	`With MCP`
Each AI app has to invent its own way to send documents, actions, and prompts. It's messy, insecure, and hard to maintain.	Apps, databases, APIs, and models can speak a **common language**.
	Tools, resources, prompts are **discoverable**, **safe**, and **interoperable**.

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MCP Core Architecture Overview

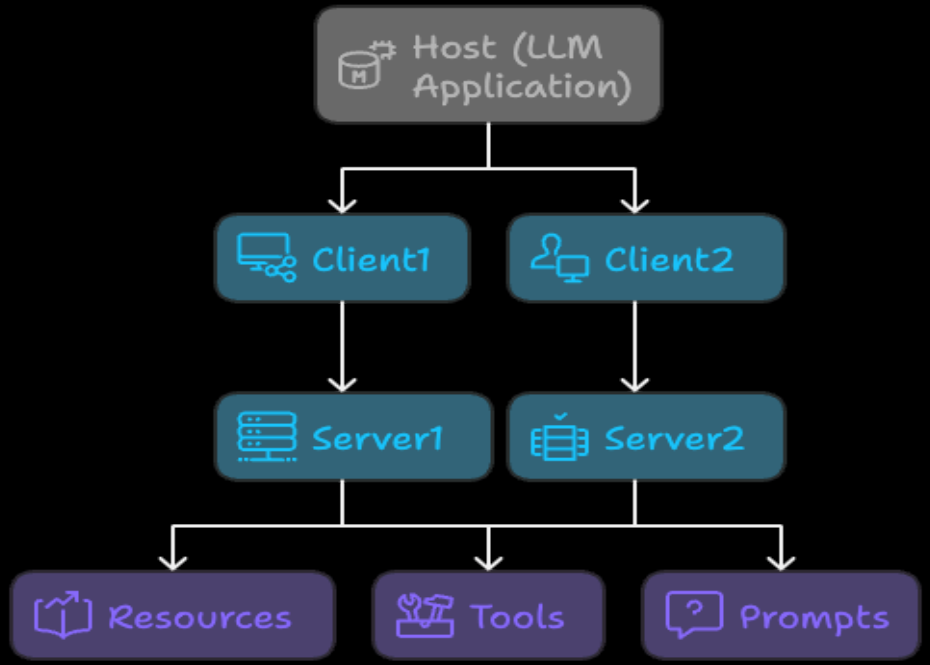
MCP is built around a **flexible, extensible client-server model** specifically optimized for LLM interactions.

🏗️ Architecture Overview

Component	Description
Host	The environment running the LLM (e.g., Claude Desktop).
Client	A bridge inside the host that manages 1:1 connections to servers.
Server	A backend that exposes **Resources**, **Tools**, and **Prompts** via the MCP protocol.

- The **Client** is embedded inside the Host (LLM app) and manages the communication.
- The **Server** is usually a Python app (or other language) that provides data and actions.

MCP Architecture Overview



MCP Core Components

The Model Context Protocol is structured around three main component layers:

- **Protocol Layer** (how we format and manage messages)
- **Transport Layer** (how we send/receive data)
- **Message Types** (the kinds of information exchanged)

Each layer is **independent and replaceable**, making MCP very modular and powerful for AI system designs.

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🚚 Transport Layer (Message Carrier)

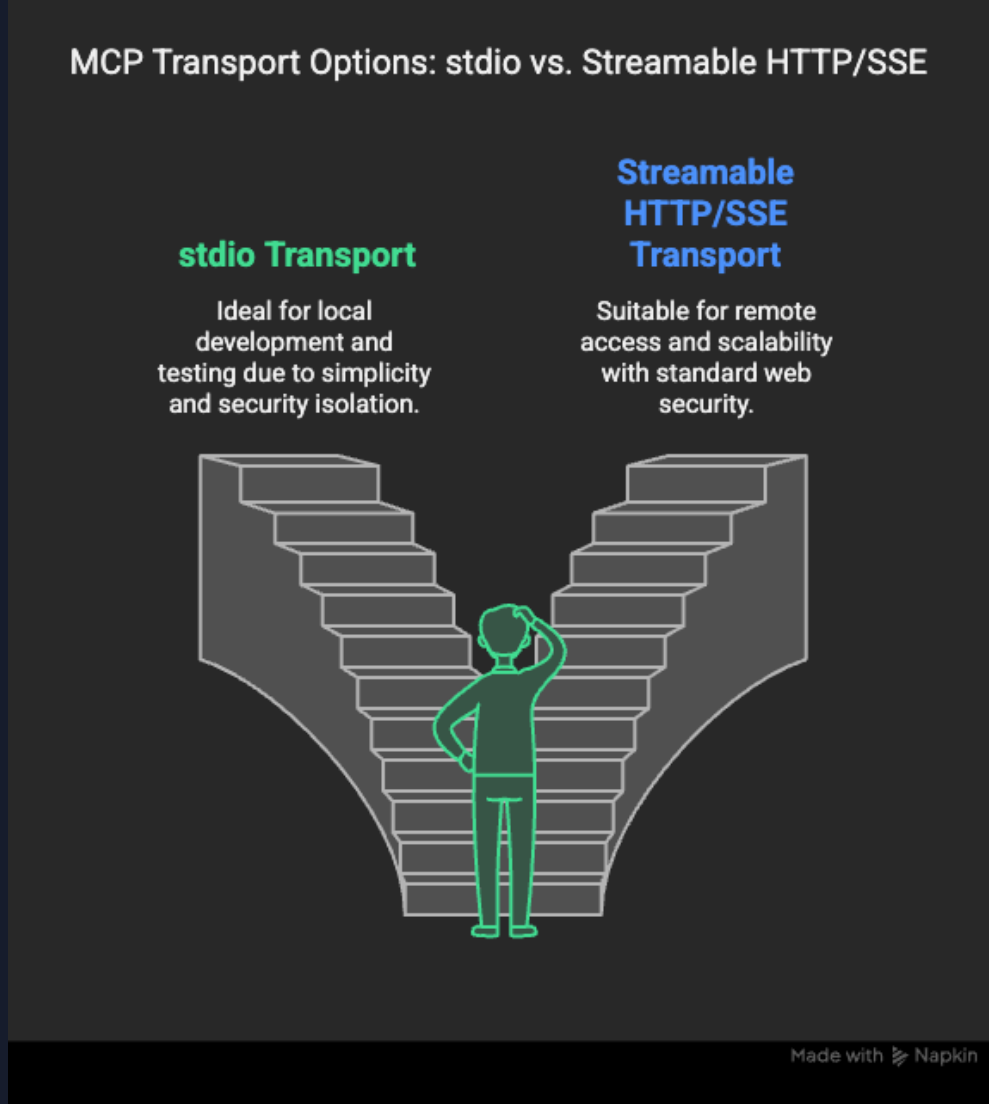
The **Transport Layer** defines **how the bytes actually travel** between Client and Server.

It **does not** care about the content – it only **moves JSON-RPC messages**.

✈️ Built-in Transport Types

Transport	Description	Use case
stdio	Standard input/output streams	Local scripts, command-line tools.
HTTP+SSE	Server-Sent Events + HTTP POST	Web apps, remote communication.

> Both use **JSON-RPC 2.0** for formatting.



JSON-RPC 2.0 Message Formats

Type	Example
Request	`{ "jsonrpc": "2.0", "id": 1, "method": "methodName", "params": { ... } }`
Response	`{ "jsonrpc": "2.0", "id": 1, "result": { ... } }`
Error	`{ "jsonrpc": "2.0", "id": 1, "error": { "code": -32600, "message": "Invalid Request" } }`
Notification	`{ "jsonrpc": "2.0", "method": "eventName", "params": { ... } }`

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MCP Connection Lifecycle

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Error Handling in Lifecycle

MCP defines standard error codes based on **JSON-RPC 2.0**:

Error Code	Meaning
`-32700`	Parse Error (Invalid JSON)
`-32600`	Invalid Request
`-32601`	Method Not Found
`-32602`	Invalid Params
`-32603`	Internal Error

Servers can define their own error codes > **-32000** for custom scenarios.

Python SDK Quickstart

🛠️ 1. Install the MCP Python SDK

```
pip install "mcp[cli]"
```

✅ This gives access to:

- MCP server libraries
- Development CLI tools (`mcp dev` , `mcp install` , etc.)

🧩 2. Create `server.py`

Here's the minimal working code:

```python
from mcp.server.fastmcp import FastMCP

# 1. Create the server
mcp = FastMCP("Demo Server")

# 2. Add a Resource
@mcp.resource("greeting://{name}")
def greet(name: str) -> str:
    """Return a personalized greeting."""
    return f"Hello, {name}! Welcome to MCP World."

# 3. Add a Tool
@mcp.tool()
def add_numbers(a: int, b: int) -> int:
    """Add two numbers together."""
    return a + b
```

✅ Highlights:

- `@mcp.resource()` exposes data (like a REST API GET).

🏃 3. Running the server locally

You can **develop and test** using the built-in CLI:

```
mcp dev server.py
```

✅ This starts a *local stdio server* and opens an **MCP Inspector UI** (in your browser) for easy testing!

If you want to run it directly:

```
python server.py
```

(But this will need a manual client to connect.)

Testing the server manually

Using `mcp` CLI:

```
mcp call-resource greeting://Alice
```

Should return:

```
{
  "text": "Hello, Alice! Welcome to MCP World."
}
```

Calling tool:

```
mcp call-tool add_numbers --args '{"a": 5, "b": 7}'
```

Should return:

```
{
  "text": "12"
}
```

📚 6. MCP Core Primitives: Resources, Prompts, Tools

Aspect	Resources	Prompts	Tools
Purpose	Provide static/dynamic context	Structure interaction	Perform actions
Control	Application-controlled	User-controlled	Model-controlled
Data	Text/Binary	Structured Messages	Action Outputs
Example	Fetch project docs	Guide code review	Create Jira Ticket

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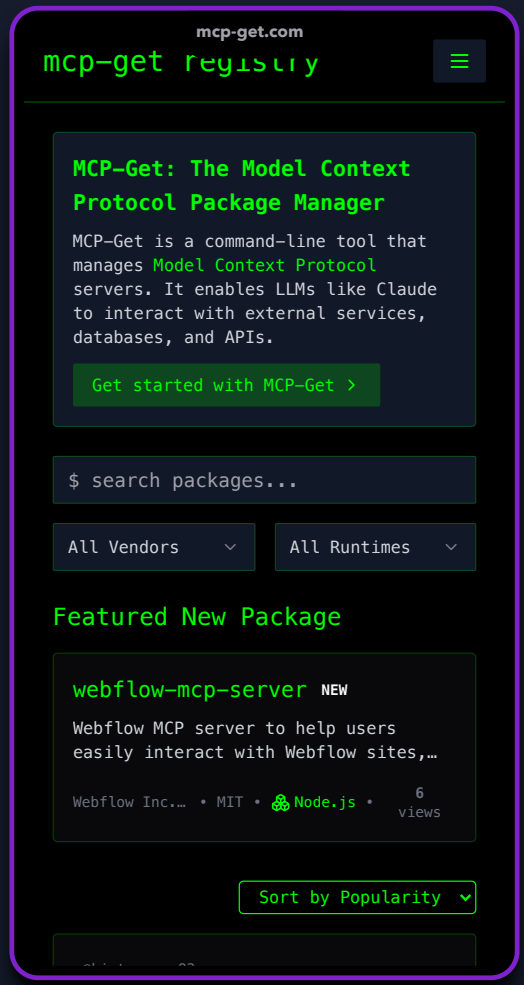
mcp-get.com
mcp-get registry
MCP-Get: The Model Context Protocol Package Manager
MCP-Get is a command-line tool that manages Model Context Protocol servers. It enables LLMs like Claude to interact with external services, databases, and APIs.
Get started with MCP-Get
$ search packages...
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